

LDM-DTI: A multimodal framework integrating pretrained language models and geometric graph networks for interpretable drug-target interaction prediction

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ABSTRACT

Accurate prediction of drug-target interactions (DTIs) is essential for speeding up the discovery of new therapeutics. Although significant progress has been made with deep learning-based approaches, considerable challenges remain in learning informative molecular representations and modeling the intricate nature of drug-target associations. To overcome these limitations, an end-to-end predictive architecture, termed LDM-DTI, is proposed. In this framework, drug and protein sequences are encoded via pretrained large language models. Specifically, ChemBERTa is utilized to derive high-dimensional semantic and structural features from SMILES strings, while ProtBERT is employed to extract contextual representations from amino acid sequences. To further incorporate spatial molecular information, a three-layer Graph Convolutional Network (GCN) and an Equivariant Graph Neural Network (EGNN) are integrated to capture both 2D topological and 3D geometric characteristics of drug molecules. Protein-level features are refined through dynamic convolutional operations and multi-head self-attention mechanisms. These representations are then fused via a Dynamic Interactive Attention Module (DIAM) to model cross-modal dependencies between drugs and targets. The proposed framework demonstrates superior predictive performance and generalizability across four public benchmark datasets, consistently surpassing ten state-of-the-art baselines. Ablation experiments are conducted to quantify the contributions of individual components, and protein-level attention maps are visualized to enhance interpretability. Overall, LDM-DTI offers a robust and interpretable solution for DTI prediction, with strong potential for accelerating structure-informed drug discovery.

1. Introduction

The accurate identification of DTIs is regarded as a critical foundation for contemporary drug discovery efforts. It provides strong support for accelerating drug discovery, improving therapeutic efficacy and reducing research and development costs. De novo drug discovery constitutes an intrinsically prolonged and resource-intensive process that progresses through a series of essential stages, beginning with the identification of disease indications and shaped by unmet clinical needs alongside evaluations of market viability. This developmental trajectory typically unfolds through sequential stages, including disease-associated

gene discovery, target validation, lead compound discovery, lead compound optimization, preclinical assessment, and multi-phase clinical trials, with regulatory authorization contingent upon comprehensive validation of both safety and therapeutic efficacy. This process typically spans 10–17 years and necessitates capital investments exceeding \$2–3 billion USD (Chu et al., 2022). Despite its inherent complexities, de novo drug discovery remains the cornerstone of the pharmaceutical industry, underpinning significant advancements in therapeutic innovation and contributing profoundly to global public health outcomes. Moreover, improvements in DTI predictive fidelity have enabled high-throughput virtual screening of extensive compound libraries, thereby

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transforming early-stage drug discovery via computational prioritization and rational target selection strategies (Jarada et al., 2020; Li et al., 2016; Sachdev & Gupta, 2019).

In recent years, computational power has increased, and large-scale biomedical datasets have become more widely available. These developments have accelerated research in DTI prediction and provided a strong basis for drug screening and repurposing (Chu et al., 2021; Liu et al., 2024; Nascimento et al., 2016). Multiple computational modeling paradigms have been proposed. These include similarity-based heuristic algorithms, matrix factorization techniques, feature engineering methods, network-based computational models, and deep learning frameworks. Each of these approaches has improved prediction efficiency and accuracy, helping to speed up the drug discovery process. Despite these advances, predicting DTIs remains challenging. Existing methods often struggle in real biological scenarios and cannot fully satisfy the requirements for precise drug design.

First, poor data quality and limited dataset size create major challenges for improving model performance. Existing DTI datasets often contain noise and inconsistent annotations, especially in low-resource scenarios. The lack of high-quality labeled samples further reduces the robustness and generalizability of predictive models (Nguyen et al., 2020). Second, drug molecules and proteins have complex 3D structures and dynamic biophysical behaviors. Traditional shallow methods that rely on static sequences or 2D representations often cannot capture these features accurately. These limitations restrict the model from identifying meaningful interaction patterns and understanding complex biological mechanisms. Recent models include MolTrans (Huang et al., 2021), which employs a Transformer architecture to capture substructural patterns and semantic relationships. However, they still struggle to model higher-order dependencies present in real DTIs. Limited generalizability remains another major challenge. Many deep learning models perform well on training data, but their accuracy often drops when applied to new compounds or unseen targets, showing poor adaptability to structural diversity. Model interpretability is another important concern. Most high-performing models rely on deep neural networks, which are considered "black-box" architectures and provide limited mechanistic understanding. This limitation reduces their usefulness in clinical validation and pharmacological studies (Zhai et al., 2022).

To address the deficiencies of current DTI prediction methods in capturing informative features and achieving robust generalization, we introduce a novel predictive framework, LDM-DTI. The model uses pretrained language models to generate informative embeddings from SMILES strings and protein sequences. This helps improve the precision of DTI modeling. Unlike traditional approaches, LDM-DTI integrates a three-layer GCN and an EGNN into the drug encoding process. The GCN captures both local and global topological features of molecular graphs, enhancing the chemical representations. The EGNN encodes three-dimensional geometric properties and symmetry information, providing richer spatial context. Protein features are extracted using the pretrained ProtBERT model. A hybrid module combining dynamic convolutional networks and Transformer-based attention is applied to refine these embeddings. As a result, the model effectively captures detailed semantic and structural information. For cross-modal integration, we design a cross-attention mechanism with dynamically learned weights. This mechanism aggregates drug and protein embeddings to produce context-aware interaction representations. The main innovations and contributions of this research can be outlined as follows:

- We propose a multimodal feature fusion framework that performs end-to-end learning. The framework covers feature extraction from drug molecules and protein sequences, cross-modal integration, and final DTI prediction. It systematically combines chemical properties, geometric structures, and biological sequence information. This comprehensive integration supports accurate prediction of DTIs.
- We integrate a dynamically weighted cross-attention mechanism to allow bidirectional interaction between drug and protein features.

This mechanism adjusts interaction weights adaptively, strengthening the semantic connection between the two. As a result, it improves prediction accuracy.

- We use multiple pre-trained models to encode features. ChemBERTa and ProtBERT, which are trained on chemical and biological sequences, are employed to generate informative embeddings for drug SMILES strings and protein sequences. This approach enables efficient characterization of molecular and protein features.
- We present a novel approach for modeling geometric features of drug molecules. The EGNN is applied to incorporate 3D geometric information into drug feature extraction for the first time. This enhances the representation of drug molecules in terms of geometric symmetry and topology and provides new insights for DTI prediction.

Section 2 then reviews the studies that are most relevant to this work. Section 3 presents the architecture of the proposed model and explains how it is implemented. Section 4 compares the performance of LDM-DTI with several baseline models. Section 5 concludes the paper and outlines possible directions for future work.

2. Related work

At present, five primary categories of DTI prediction methods are recognized: similarity/distance-based methods, deep learning methods, matrix decomposition-based methods, feature-based methods, and network-based methods. A detailed description of each method is presented below.

Similarity- or distance-based methods rely on the principle that compounds with similar structures are likely to share biological activities and interaction patterns. These methods usually start by extracting structural, physicochemical, and functional descriptors for both drugs and targets, and then compute similarity scores from their feature vectors (Olayan et al., 2018). Next, a predictive model is built using known interaction data to estimate how likely unseen drug-target pairs are to interact. To improve the accuracy and expressiveness of similarity-based models, several optimization methods have been proposed. For example, A composite scoring scheme (Perلمان et al., 2011) integrated drug-drug and gene-gene similarity matrices to predict drug-gene associations. By combining different similarity types within a logistic regression model, it produced more robust predictive scores. Another methods, FGS (Liu et al., 2023) used a fine-grained selective similarity integration mechanism to capture local structural patterns with a weighted consistency matrix. Building upon this, Neighborhood interaction information (NII) was added to the BLM (Shi et al., 2015) framework to better handle unstructured data. To deal with incomplete interaction annotations, the concept of "super-targets" was introduced to represent groups of functionally similar targets, helping the model generalize more effectively. Despite their proven utility, similarity-based models are limited by the quality of similarity metrics, incomplete data, and susceptibility to noise. Therefore, improving predictive fidelity and model generalization often requires careful selection of similarity functions and integration with more expressive learning architectures.

Recently, deep learning has been widely applied in drug discovery, particularly for DTI prediction, providing significant advantages by addressing key limitations and reducing representational degradation. DeepDTA (Ozturk et al., 2018) was one of the first models for DTI prediction. It learns drug and protein features using Convolutional Neural Networks (CNNs) and multilayer perceptrons. Later, MDCT-DTA (Zhang et al., 2022) introduced a deep learning approach combining multi-scale graph diffusion, CNNs, and a Transformer module to enhance affinity prediction. It effectively models complex molecular and cross-modal interactions. MSN-DTA (Wang et al., 2025) was proposed to improve drug representation accuracy and model generalization by integrating multi-scale GNNs with adaptive node feature learning. In addition, DTIAM (Lu et al., 2025) was developed with self-supervised pre-training

to extract drug and target features from large unlabeled datasets, allowing the model to capture molecular substructures and contextual meaning. Although deep learning has brought remarkable progress in DTI prediction, several issues remain. In particular, dependence on large datasets and limited interpretability continue to restrict performance. Future studies may explore integrating deep learning with other analytical methods to improve both prediction accuracy and interpretability.

Matrix decomposition-based methods have been widely adopted as an effective computational strategy in DTI prediction. These methods represent known DTIs as a two-dimensional matrix. Each entry shows whether a specific interaction exists. The interaction matrix and auxiliary drug and target features were then used to infer unknown interactions through matrix factorization or completion techniques. Two matrix decomposition algorithms introduced (Ezzat et al., 2017) predicted interactions for "novel drugs" and "novel targets". They included intermediate associations and assigning interaction scores to network edges. SPLCMF (Xia et al., 2019) used a stepwise sample selection strategy to identify interactions with unseen targets. The method also integrated low-order weighted collaborative matrix factorization. In addition, it embedded drug-target association networks into a least squares framework to obtain simpler, low-dimensional embeddings. A hybrid DTI prediction model (Sajadi et al., 2022) was proposed. It included preprocessing of the interaction matrix and combined biological data from different sources. By using drug and target similarity matrices, this approach reduced matrix sparsity. Even though it showed better robustness than traditional techniques, the fast growth and diversity of drug-related data exceeded the modeling capacity of matrix-based frameworks and traditional analytical methods.

Network-based methods treat drugs and targets as parts of a heterogeneous network. Graph-based algorithms are then used to predict potential DTIs. This approach allows the discovery of previously unobserved interactions by revealing hidden relational patterns. Graph neural networks (GNNs) have been extensively used to model heterogeneous graphs. From these graphs, high-order topological representations are extracted to improve predictive performance (Li et al., 2023; Manoochehri et al., 2020; Yu et al., 2021; Zhao et al., 2024). ZeroBind (Wang et al., 2023) introduced a GNN combined with a weakly supervised subgraph information bottleneck module. It generated task-specific embeddings for molecular and protein graphs and identified potential binding pockets in protein structures. VEAGDTI (Zhang et al., 2023) captured deep interaction representations by integrating heterogeneous networks. Mutual-DTI (Wen et al., 2023) used a Transformer encoder to merge protein sequence-level interaction features and better model inter-sequence dependencies. AOPEDF (Zeng et al., 2020) preserved arbitrary-order proximity through compact embedding learning from a heterogeneous network of drugs, proteins, and diseases. GSRF-DTI (Zhu et al., 2024) combined diverse molecular and target information to construct a DTI network and explicitly captured the influence of these associations on predictive performance. A multimodal drug-target knowledge graph (He et al., 2025) was constructed, and essential topological representations were extracted using a recursive model. DMHGNN (Ning et al., 2025) integrated *meta*-path reasoning, denoising autoencoders, and a multi-channel GNN. This allowed efficient representation learning on a bi-diual multiview heterogeneous graph. Despite their effectiveness, network-based DTI prediction models still rely heavily on prior knowledge about drugs, targets, and known interactions.

Feature-based DTI prediction approaches have received considerable attention. They use machine learning techniques to extract useful representations from available drug and protein data and build predictive models. A computational framework (Huang et al., 2018) was introduced, combining a random tree algorithm with protein-amino acid descriptors. In this method, protein sequences were encoded into pseudo-substitution matrices to retain evolutionary information, while drug molecules were represented using fingerprint-based feature vectors to capture substructural patterns. PCNN-DTA (Chen et al., 2023) was

proposed to incorporate a feature pyramid network. This network fuses multi-level features extracted from convolutional layers of different depths. The hierarchical fusion helps preserve fine-grained structural information and improves prediction accuracy. Potential DTIs are identified by jointly modeling protein sequences and molecular structures. Despite these improvements, such methods still face limitations. The absence of three-dimensional structural data for membrane proteins restricts their ability to capture detailed spatial characteristics.

The use of language modeling for drug and protein feature extraction has brought new momentum to DTI prediction research. It allows for more expressive molecular representations. Large language models, such as BERT (Devlin et al., 2019) originally transformed natural language processing. Then, these principles were adapted for protein modeling, leading to protein embedding models including ESM (Lin et al., 2023), ProtBERT (Elnaggar et al., 2022), and ProteinBERT (Brandes et al., 2022). For drug representation, pre-trained models like ChemBERTa, ChemGPT, and MolFormer have been widely used. LGDTI (Zhao et al., 2021) framework integrated language model capabilities with graph representation learning techniques. DeepWalk (Perozzi et al., 2014) used GNNs to capture local molecular structures while utilizing graph embedding techniques to extract higher-order neighborhood information. DTI-LM (Khandakar Tanvir et al., 2024) employed ChemBERTa and ProtBERT to encode SMILES strings and protein sequences. These embeddings are combined with GNNs to capture structural features. However, the embeddings remain independent, lacking explicit modeling of local neighborhood structures. This constrains the joint expression of drug-target features. Including neighborhood information has been shown to substantially improve drug and target representations (Wan et al., 2019). Most existing methods focus mainly on comparing different language model variants and do not evaluate them comprehensively against other mainstream approaches. Current progress in DTI prediction largely concentrates on refining drug and protein encoders or optimizing classifier performance (Hua et al., 2025; Tsubaki et al., 2019; Yu et al., 2024). Future research will explore encoder synergy mechanisms and cross-modal feature fusion strategies to enhance model generalization and interpretability.

3. Materials and methods

3.1. Datasets

The proposed model was evaluated in comparison with nine baseline methods across four public datasets: BindingDB, BioSNAP, *Human*, and *C. elegans*. For the experiments, A refined and less-biased variant of the BindingDB dataset (Bai et al., 2021) was employed. It contains experimentally validated binding affinity data between small-molecule drugs and proteins, comprising 14,643 drugs, 2,623 proteins, and 49,199 interaction records. The dataset integrates data from literature and experiments, reflecting real-world DTIs. The BioSNAP dataset, derived from the DrugBank database (Wishart et al., 2008), includes 4,510 drugs and 2,181 proteins. Positive interactions are experimentally verified, while negative samples are generated by randomly pairing drugs and proteins with no reported interactions. This ensures a balanced number of positive and negative samples. Similarly, the *Human* dataset (Zheng et al., 2020) contains 2,726 drugs, 2,001 proteins, and 5,997 interactions. Negative samples are constructed from drug-protein pairs with no known interactions to maintain balance. The *C. elegans* dataset focuses on the model organism *Caenorhabditis elegans*. It includes 1,716 drugs, 1,856 proteins, and 7,509 interaction records, offering a valuable reference for studying complex biological systems. Overall, the model was evaluated comprehensively on these four datasets. They cover a wide range of drug and protein types, interaction scales, and biological contexts. The datasets provide data relevant to humans as well as model organisms, and carefully curated negative samples to reduce potential biases during training and evaluation. The detailed information of each dataset is presented in Table 1.

Table 1

Summary of the datasets used in this study.

Dataset	Drug	Protein	Interaction
BindingDB	14,643	2623	49,199
BioSNAP	4505	2181	27,464
Human	2726	2001	5997
<i>C.elegans</i>	1716	1856	7509

3.2. Overview of LDM-DTI framework

In this study, DTI modeling is enhanced by combining information from language models and graph-based structural encoders. We propose a multi-modal deep learning model called LDM-DTI, as shown in Fig. 1. As depicted in Fig. 1(A), the data processing module employs ChemBERTa and ProtBERT to encode drug and protein sequences. This produces atom-level embeddings for drugs and amino acid-level embeddings for proteins. The embeddings are then passed through their respective feature encoders. The protein feature extraction module, illustrated in Fig. 1(B), incorporates a Dynamic Convolutional Neural Network (DCNN) with positional encoding, residual connections, and a feed-forward network to effectively extract and refine protein sequence features. As shown in Fig. 1(C), the drug feature extraction module employs a three-layer graph GCN to capture the structural characteristics of molecular graphs by integrating the adjacency and degree matrices. In addition, EGNN is integrated to strengthen the model capability in capturing topological characteristics. The DIAM, as illustrated in Fig. 1(D), is employed to perform feature fusion of the drug and protein representations following feature extraction. Each modality is processed via multi-head self-attention and dynamically weighted to facilitate effective multi-modal integration. A joint feature vector is subsequently generated through a max-pooling operation. Finally, as shown in Fig. 1(E), the prediction and evaluation module inputs the fused vector into a multi-layer perceptron (MLP) to estimate the likelihood of a DTI. A variety of evaluation metrics and DTI network analysis are employed to assess the predictive performance of the proposed model. All source code used in this study is publicly accessible at <https://github.com/YuBinLab-QUST/LDM-DTI/>.

3.3. Drug representation learning

To generate drug representations, SMILES sequences were encoded using ChemBERTa (Chithrananda et al., 2020), a RoBERTa-based model designed for chemical language modeling. ChemBERTa applies a word embedding approach to convert each character or substructure in a SMILES string into a dense vector. For instance, each token, such as the first character in the sequence, is mapped to an embedding vector that reflects its contextual and structural significance within the molecule.

$$\vec{e}_i = E(c_i) \quad (1)$$

The hidden dimension of the ChemBERTa model was configured to 512 to balance computational efficiency and the capacity to represent diverse chemical features. Additionally, drug SMILES sequences were standardized by truncation or padding to a fixed length of 512 tokens, in accordance with the maximum input length supported by the ChemBERTa encoder.

3.4. Protein representation learning

ProtBERT was employed to encode individual amino acids within the protein sequence. This model, based on the Transformer architecture, comprises a core formed by multiple stacked encoder layers. It uses a predefined amino acid vocabulary to map each amino acid to a unique index. These indices are then transformed into low-dimensional vectors through the embedding layer, mapping the amino acids into a low-dimensional vector space. For the i -th amino acid a_i in the protein

sequence, a corresponding vector representation is obtained by applying the embedding function E . This process is formally defined as:

$$\vec{e}_i = E(a_i)N = \text{Pooling}(X') = \sum_{i=1}^N x'_i \quad (2)$$

In this study, the hidden dimension of the ProtBERT model was configured to 1024 to enable comprehensive extraction of protein features while ensuring computational efficiency. During preprocessing, protein sequences were standardized to a maximum length of 1200 tokens through truncation or padding. For sequences shorter than 1200 residues, [PAD] tokens were appended, and an attention mask was generated to distinguish valid amino acid positions from padding tokens. During the Transformer encoding process, this mask was applied to the self-attention mechanism to prevent the padded positions from influencing the computation, thereby ensuring that only meaningful residues contribute to the contextual embeddings. The normalization ensured that the inputs were compatible with the model and minimized information loss.

3.5. Drug feature encoder

Prior to feature extraction via a GCN (Kipf et al., 2017) each drug molecule is converted into a graph-structured representation, where atoms are represented as vertices and chemical bonds as edges. To obtain informative node features, the pre-trained ChemBERTa model is employed to encode each vertex, generating the embedding representation of the drug.

A three-layer GCN is employed in this study for drug feature extraction. At the i -th layer ($i = 1, 2, 3$), each node receives information from its neighboring nodes. Each node k aggregates feature information from its adjacent nodes $l \in N(k)$, where $N(k)$ denotes the set of neighbors of node k . The message propagation is mathematically expressed as follows:

$$m_k^{(i)} = \sum_{l \in N(k) \cup \{k\}} \frac{1}{\sqrt{\hat{d}_k \hat{d}_l}} H_l^{(i-1)} W^{(i)} \quad (3)$$

where k is the degree of node $\hat{d}_k$ after considering self-loops. When $i = 1$, which represents the initial node feature matrix. $W^{(i)} \in R^{F_{i-1} \times F_i}$ is the learnable weight matrix of the i -th GCN layer, and F_i is the output feature dimension of the i -th GCN layer.

Following the aggregation of messages from neighboring nodes, the resulting features are transformed through a nonlinear operation using the ReLU activation function.

$$H^{(i)} = \text{ReLU}(M^{(i)}) \quad (4)$$

where $M^{(i)}$ denotes the matrix of all nodes after message aggregation at layer i and $H^{(i)} \in R^{N \times F_i}$ represents the output feature matrix of the GCN at layer i .

After three layers of GCN processing, each node obtains a feature representation that incorporates information from its neighboring nodes. To derive a feature representation for the entire drug molecule, a global pooling operation is applied to the features of all nodes.

$$M = \text{Pooling}(H^{(3)}) \quad (5)$$

where $M \in R^{F_3}$ is the global feature vector of the entire drug molecule.

Drug feature extraction using EGNN (Garcia Satorras et al., 2022) begins by representing the drug molecule as a graph, where atoms are modeled as nodes and chemical bonds as edges. The initial node features are represented by drug embedding vectors obtained through ChemBERTa encoding, with the node feature matrix denoted as $X \in R^{N \times F}$, where N represents the number of atoms and F denotes the dimensionality of the embedding vector. In addition, spatial coordinates $R =$

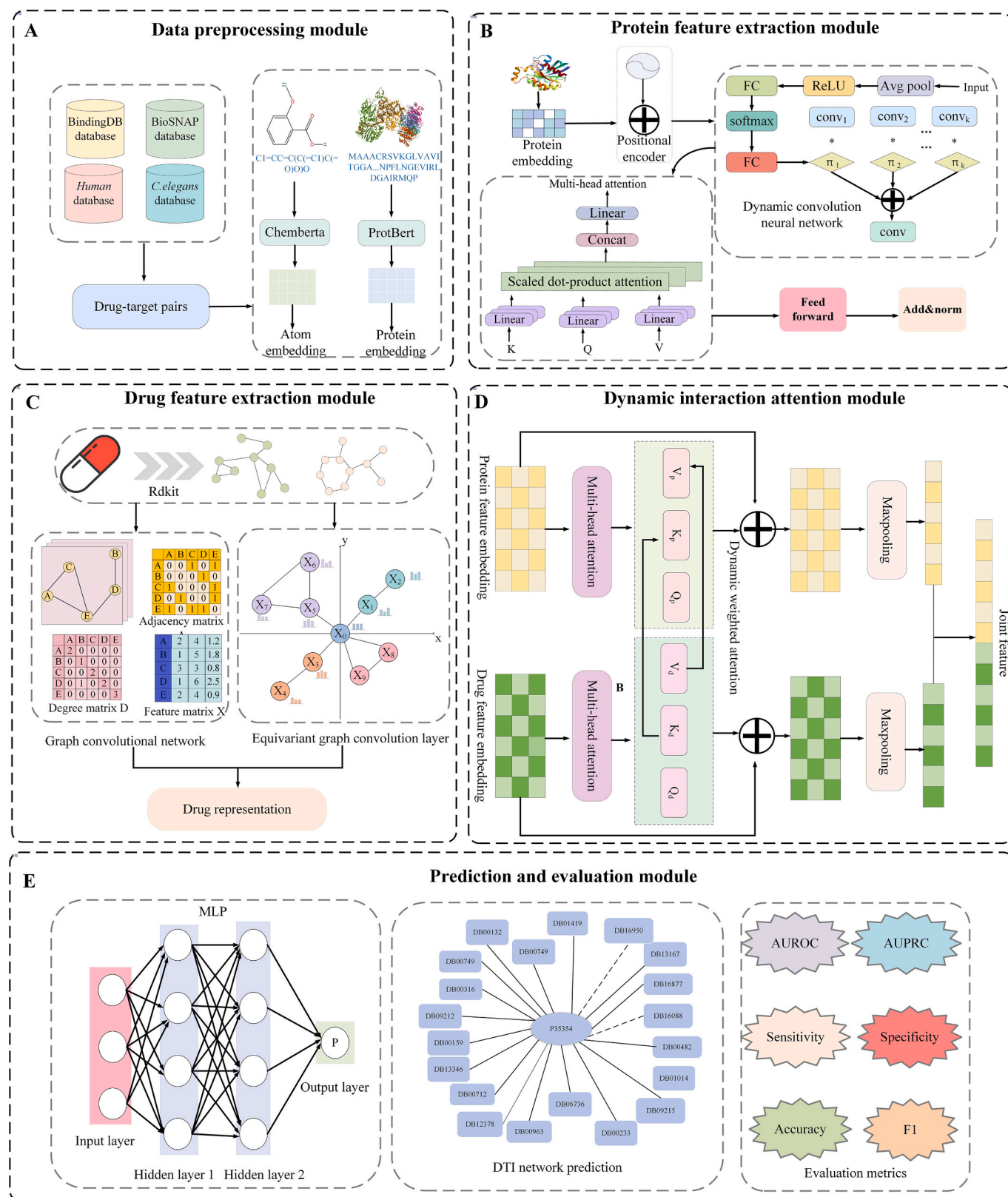


Fig. 1. The comprehensive framework of LDM-DTI. (A) Data preprocessing module. (B) Protein feature extraction module. (C) Drug feature extraction module. (D) Dynamic interaction attention module. (E) Prediction and evaluation module.

$[r_1, r_2, \dots, r_N]^T \in \mathbb{R}^{N \times 3}$ are assigned to each node to describe atom positions in 3D space. Specifically, the atomic coordinates were derived by hydrogenating the SMILES molecular formula, generating multiple three-dimensional conformations, and optimizing them using the UFF

force field. This process resulted in the three-dimensional coordinates of each atom, as computed by RDKit, thereby supplying the EGNN model with accurate atomic spatial information.

A message passing mechanism is typically employed by EGNNs to

update node features. At each message passing layer, the feature representation of a node is updated according to the attributes and relative positional information of its neighboring nodes. For node i the message m_{ij} it receives from its neighboring node j is computed based on the features x_i, x_j of the two nodes and their relative position $r_{ij} = r_j - r_i$. The message computation function is defined as follows:

$$m_{ij} = \phi(x_i, x_j, r_{ij}) \quad (6)$$

where ϕ is a learnable function.

The messages received by node i from all neighboring nodes are aggregated to obtain the aggregated message m_i . In this study, summation is employed for message aggregation and is computed as follows:

$$m_i = \sum_{j \in \mathcal{N}(i)} m_{ij} \quad (7)$$

Upon receiving the aggregated message m_i , the features of node x_i are updated accordingly. The specific update formula is given as follows:

$$x'_i = \psi(x_i, m_i) \quad (8)$$

where ψ denotes a learnable function, which can be implemented using a multilayer perceptron.

To maintain SE(3)-equivariance, EGNNs must update not only node-level features but also atomic coordinate representations. Specifically, for any pair of atoms i and j within a molecular graph, their relative positional vector r_{ij} and corresponding Euclidean distance d_{ij} . The relative position is derived using the following formulation:

$$r_{ij} = r_j - r_i \quad (9)$$

The Euclidean distance is computed as follows:

$$d_{ij} = \|r_{ij}\|_2 \quad (10)$$

The relative positional vector and Euclidean distance serve as critical inputs for subsequent edge feature computation and coordinate refinement.

Edge features represent the interactions between atoms, containing information such as the interatomic distance and bond type. For each edge (i, j) , the edge feature vector e_{ij} is constructed by taking the features of nodes i and j , denoted as x_i and x_j , and the interatomic distance d_{ij} as inputs.

$$e_{ij} = [x_i, x_j, d_{ij}] \quad (11)$$

The edge feature vectors are processed by an edge multilayer perceptron to yield the edge features m_{ij} :

$$m_{ij} = \text{EdgeMLP}(e_{ij}) \quad (12)$$

The computation of edge features enhances the capture of atomic interactions and provides more precise information for coordinate updates.

After obtaining the edge features, the weighted influence of these features on the nuclear coordinates can be computed and applied to update the coordinates. The specific coordinate update formula is as follows:

$$r_i^{\text{new}} = r_i + \sum_j w_{ij} r_{ij} \quad (13)$$

where w_{ij} is the weight obtained through an MLP network, indicating the degree of influence of edge (i, j) on the coordinate update process. In this manner, the coordinates of a node can be dynamically updated based on the information from its neighboring nodes, thereby more accurately reflecting the spatial structural changes of drug molecules.

To derive a comprehensive representation of the entire drug molecule, feature information from all atomic nodes is integrated. In the

present study, a summation-based pooling strategy is adopted, where the feature vectors of individual nodes are added together to form a unified global feature vector for the molecule:

$$N = \text{Pooling}(X') = \sum_{i=1}^N x'_i \quad (14)$$

where X' is the feature matrix of all nodes after multilayer processing.

Finally, the node features $N \in R^{N \times F_2}$ output from the EGNN are concatenated with the feature matrix $M \in R^{N \times F_1}$ obtained from the three-layer GCN extraction to form a comprehensive feature representation matrix $P \in R^{N \times (F_1 + F_2)}$, where N denotes the number of nodes, and F_1 and F_2 represent the output feature dimensions, as given by the following formula:

$$P = [M \quad N] \quad (15)$$

3.6. Protein feature encoder

Dynamic convolution and multi-head self-attention frameworks are utilized as protein feature encoders to effectively capture both local and global characteristics of protein sequences within the proposed approach. Protein embedding vectors generated by ProtBERT are provided as input to the encoder, as defined below.

$$E = [e_1, e_2, \dots, e_L] \in R^{L \times d} \quad (16)$$

where L is the length of the protein sequence and d is the embedding dimension. Subsequently, E passes through positional encoding, dynamic convolution, multi-head self-attention, and a feed-forward neural network sequentially to ultimately obtain the deep feature representation of the protein.

Since Transformer architecture lacks inherent modeling of sequential positions, learnable positional encodings are incorporated into input embeddings to compensate for this limitation:

$$E_p = E + P, P \in R^{L \times d} \quad (17)$$

where P is learned during training to explicitly provide order information in the embedding space. The updated representation E_p is then used as input to the dynamic convolutional layer.

Local patterns are extracted from protein sequences, dynamic convolution is employed for feature extraction. Unlike conventional CNNs with fixed kernels, the dynamic convolution mechanism adaptively adjusts its filters according to input features, enabling context-aware receptive fields that better capture local sequence variations and residue dependencies. This adaptability allows the model to learn subtle, position-specific biochemical patterns that static CNNs may overlook, resulting in more informative and biologically meaningful protein representations. Let $W_{\text{conv}} \in R^{k \times d \times d}$ denote the convolution kernel and k the convolution window size; the resulting representation after convolution is given by:

$$H_{\text{conv}} = \text{ReLU}(E_p * W_{\text{conv}} + b) \quad (18)$$

where $b \in R^d$ is the bias term, $*$ denotes the one-dimensional convolution operation, and $H_{\text{conv}} \in R^{L \times d}$ represents the output after local feature extraction. This feature representation is passed to the multi-head self-attention layer to capture global dependencies.

Following extraction of local feature representations, a multi-head self-attention mechanism is applied to capture long-range dependencies among amino acid residues within the sequence. Specifically, the query, key, and value matrices are first generated through linear transformations:

$$Q = H_{\text{conv}} W_Q, K = H_{\text{conv}} W_K, V = H_{\text{conv}} W_V \quad (19)$$

where $W_Q, W_K, W_V \in R^{d \times d'}$ is a trainable weight matrix. The scaled dot-

product attention is then computed as follows:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d^*}}\right)V \quad (20)$$

The model is designed to capture diverse patterns. It employs a multi-head self-attention mechanism with eight attention heads. This setup enables the model to learn several interaction subspaces at the same time and to capture contextual dependencies among amino acid residues.

$$H_{attn} = \text{Concat}(A_1, \dots, A_h)W_O \quad (21)$$

where h denotes the number of attention heads, and $W_O \in R^{(h \cdot d^*) \times d}$ represents the projection matrix. The resulting representation H_{attn} is first combined with the input through a residual pathway and subsequently refined using layer normalization.

In order to enhance training robustness, residual integration and normalization are performed on the output H_{attn} of the MHSA module:

$$H_{norm1} = \text{LayerNorm}(H_{conv} + H_{attn}) \quad (22)$$

Subsequently, a feedforward neural network is utilized to further derive high-level feature representations.

$$H_{ffn} = \text{ReLU}(H_{norm1}W_1 + b_1)W_2 + b_2 \quad (23)$$

where $W_1 \in R^{d \times d_{fin}}, W_2 \in R^{d_{fin} \times d}$ are trainable weight matrices, and b_1 and b_2 are the corresponding bias vectors. Ultimately, layer normalization is applied once more to stabilize output.

$$V = \text{LayerNorm}(H_{norm1} + H_{ffn}) \quad (24)$$

where $V \in R^{L \times d}$ denotes the final output vector of the protein feature encoder.

3.7. Dynamic weighted fusion module

A feature interaction module leveraging dynamic weighting is developed to improve representations of DTI. Through the integration of multi-head attention and a dynamic weighting strategy, this module effectively models these interactions, resulting in a high-level feature representation of DT pairs.

At preceding stages, the drug and protein features have been encoded. The drug feature matrix is denoted as $P \in R^{B \times N_d \times d}$, and the protein feature matrix as $V \in R^{B \times N_p \times d}$, where B represents the batch size, N_d and N_p denote the number of features for the drug and protein, respectively, and d is the dimension of each feature. At this stage, P and V contain the high-dimensional feature representations of drugs and proteins, which are subsequently used as inputs to the feature cross-fertilization module. In this module, a multi-head attention mechanism with eight attention heads is employed to enable the model to jointly capture diverse interaction subspaces between drugs and proteins.

Drug and protein features are computed through a multi-head attention mechanism to enhance their representation. For the drug features p and protein features V , queries, keys, and values are generated via linear transformations:

$$Q_p = W_q P, K_p = W_k P, V_p = W_v P \quad (25)$$

$$Q_v = W_q V, K_v = W_k V, V_v = W_v V \quad (26)$$

where $W_q, W_k, W_v \in R^{d \times d}$ are the learnable parameter matrices. This process enables the model to capture the characteristic relationships within the drug and protein features.

We use a dynamic attention mechanism with parameterized weights to better characterize the interaction patterns between drugs and proteins. The key matrix is adjusted using trainable parameters. This allows the model to extract more informative and discriminative interaction

features. For drug representations, the revised key matrix is expressed as follows:

$$K'_p = \alpha I + K_p K_p^T \quad (27)$$

Similarly, for protein characterization:

$$K'_v = \alpha I + K_v K_v^T \quad (28)$$

where α represents a learnable dynamic weight, I denotes the identity matrix, $K_p K_p^T$ and $K_v K_v^T$ represent the autocorrelations of the drug and protein bond matrices, respectively.

The final weighted attention matrix is computed using a pseudo-inverse operation to obtain stable attention weights.

$$A_p = K_p'^+ V_p \quad (29)$$

$$A_v = K_v'^+ V_v \quad (30)$$

where $K_p'^+$ and $K_v'^+$ denote the pseudo-inverses of K'_p and K'_v , respectively. A_p and A_v denote the interactions between the drug and protein feature representations.

A bidirectional interactive updating mechanism is employed, in which drug and protein features are jointly updated through interaction to enhance their feature representations.

$$P' = P + A_p, V' = V + A_v \quad (31)$$

where A_p and A_v denote dynamically weighted attention matrices, respectively, so drug and protein features are continuously refined during the interaction to yield more accurate representations.

After feature fusion interactions, salient feature information is extracted via a global max pooling layer applied along the sequence dimension (dimension = 1) to generate the final high-level feature representations of drugs and proteins. Specifically, for the fused feature matrices of drugs $P' \in R^{L_d \times 128}$ and proteins $V' \in R^{L_p \times 128}$, where L_d and L_p denote the number of atoms and residues, respectively, the pooling operation selects the maximum activation across each of the 128 feature channels, resulting in fixed-size vectors of 1×128 .

$$V(d) = \text{Maxpool}(P') \quad (32)$$

$$V(p) = \text{Maxpool}(V') \quad (33)$$

where $V(d)$ and $V(p)$ denote the high-level feature representations of drugs and proteins during the interaction process, respectively, which capture critical information about drug-protein pairing.

Eventually, concatenation of drug and protein features is performed to generate the final interaction feature vector:

$$F = [V(d), V(p)] \in R^{2d} \quad (34)$$

3.8. MLP prediction model

DTI prediction is carried out by feeding the interaction feature vectors into an MLP, followed by successive nonlinear transformations to generate the final output. The MLP is composed of an input layer, two hidden dense layers, and an output layer. Specifically, its computational process is formulated as follows:

The fused feature vector serves as the input $F \in R^{2d}$ to the MLP. Nonlinear representation is enhanced through a two-layer fully connected network, where ReLU functions are applied following each layer.

$$H_1 = \text{ReLU}(W_1 F + b_1) \quad (35)$$

$$H_2 = \text{ReLU}(W_2 H_1 + b_2) \quad (36)$$

where $W_1 \in R^{h_1 \times 2d}, b_1 \in R^{h_1}$ represents the learnable parameter of the first hidden layer, and h_1 denotes the number of neurons in this layer.

Table 2
Detailed configurations of model modules for DTI prediction.

Module	In_dim	Hidden_dim	Out_dim
GCN	75	(128,128,128)	128
EGNN	128	128	128
DCNN	128	128	128
Transformer	128	128	128
DIAM	128	128	128
MLP	256	256	128

Similarly, $W_2 \in R^{h_2 \times h_1}$, $b_2 \in R^{h_2}$ represents the learnable parameter of the second hidden layer, and h_2 denotes the number of neurons in this layer.

The output layer is activated using a sigmoid function to generate probabilistic outputs.

$$\hat{y} = \text{sigmoid}(W_{\text{out}}H_2 + b_{\text{out}}) \quad (37)$$

where $W_{\text{out}} \in R^{1 \times h_2}$, $b_{\text{out}} \in R$ is the learnable parameter of the output layer. The predicted value $\hat{y}$ represents the probability of interaction between the drug and the protein. If $\hat{y} \geq 0.5$, an interaction is deemed to occur; otherwise, no interaction is assumed.

3.9. Training and loss

Given the binary nature of the classification task, the binary cross-entropy loss function is adopted in this work:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (38)$$

where B denotes the batch size, y_i represents the true label (0 or 1), and $\hat{y}_i$ indicates the interaction probability predicted by the model.

4. Experiments

4.1. Experiment setting

Experimental evaluations were carried out on a pi7.4xlarge.4 server equipped with a 16-core processor, 64 GB of memory, and an NVIDIA A10 GPU with 24 GB of video memory. The software environment used Python 3.8.3, and the model was implemented in PyTorch (v2.0.0 + cu117). Each dataset was trained for over 100 epochs with a batch size of 16. Data loading used 8 worker threads to improve efficiency. The learning rate started at 1e-3 and was reduced during training through a decay strategy, halving every 25 epochs with a decay factor of 0.5. Weight decay was set to 1e-3 to help prevent overfitting. We used a fixed random seed of 42 was used to ensure reproducibility and consistent data shuffling. Each dataset was evaluated over five independent runs. For each run, we select the model with the highest AUROC on validation set for final testing and recorded corresponding metrics. Table 2 summarizes the configuration of each module in the LDM-DTI framework, including input, hidden, and output dimensions. Algorithm 1 provides the pseudocode of LDM-DTI.

Algorithm 1: LDM-DTI Prediction Framework	
Input: FeatureVecs: pretrained interaction-level features DrugData: {Graph G_d , Feature X_d , Coordinates C_d } ProteinData: {Features X_p , Mask M_p }	
Output: Predicted DTI score	
1. Step 1: Drug Graph Encoding via GCN	
2. $V_{gc} \leftarrow \text{DrugGCN}(G_d)$	
3. Step 2: Drug Coordinate Encoding via EGNN	
4. $V_{egnn} \leftarrow \text{DrugEGNN}(X_d, C_d)$	
5. Step 3: Drug Feature Fusion	
6. $V_d \leftarrow \text{Concatenate}(V_{gc}, V_{egnn})$	
7. $V_d \leftarrow \text{WeightedFusion}(\text{FeatureVecs}, V_d)$	

(continued on next column)

(continued)

Algorithm 1: LDM-DTI Prediction Framework	
8. Step 4: Protein Feature Extraction	
9. $V_p \leftarrow \text{ProteinEncoder}(X_p, M_p)$	
10. Step 5: Drug-Target Fusion via Cross-Attention	
11. $\text{FusedVec} \leftarrow \text{DynamicFusion}(V_d, V_p)$	
12. Step 6: Prediction via MLP	
13. $\text{Result} \leftarrow \text{MLPClassifier}(\text{FusedVec})$	
14. return Result	
Auxiliary Function Descriptions:	
• DrugGCN: Applies multi-layer GCN on the drug graph G. • DrugEGNN: Applies EGNN to drug features and 3D coordinates. • ProteinEncoder: Extracts protein features via pretrained LM and DCNN/Transformer. • DynamicFusion: Uses cross-attention to fuse drug and protein features. • MLPClassifier: Predicts interaction score using a multilayer perceptron.	

4.2. Baselines

For comparison, ten baseline methods were selected in this study, categorized into three main groups: traditional machine learning classifiers, including SVM, RF, KNN, and LR; graph-based models such as GraphDTA, DrugBAN, IFDTI, and BINDTI, which utilize Graph Representation Learning through GCNs and CNNs; and Transformer-based models, including TransformerCPI and MolTrans. Descriptions of these baseline methods are provided below.

SVM (Cortes & Vapnik, 1995): The primary function of SVM is to classify data by learning decision boundaries in high-dimensional feature spaces. It inputs drug and protein feature vectors and generates a discriminant function to classify unseen samples.

RF (BREIMAN, 2001): RF, an ensemble learning method, combines outputs from multiple decision trees trained on randomly sampled data and features to improve generalization and control overfitting. In this study, the forest was composed of 100 trees.

KNN (Guo et al., 2003): KNN is a classical instance-based algorithm belonging to the family of supervised learning methods. In the DTI framework, drugs and targets are each represented by feature vectors. Similarity between the query drug-target pair and all known pairs in the training set was quantified using Euclidean distance. To ensure consistency, k was uniformly set to 5.

LR (Friedman et al., 2000): Logistic regression applies a sigmoid transformation to a weighted sum of input features, resulting in a probability suitable for binary classification. In DTI prediction, relevant features are first selected, and the output of their linear combination is then transformed into a probability between 0 and 1 using the logistic function.

GraphDTA (Nguyen et al., 2021): The drug is represented as a molecular graph and encoded using a GNN. The protein sequence is processed by a convolutional neural network. The resulting representations are then combined and passed through a series of fully connected layers to predict the drug-target affinity.

DrugBAN (Bai et al., 2023): Drug molecular graphs and protein sequences are processed using a deep attention-based model with domain adaptation. Conditional domain adversarial learning is applied to reduce discrepancies between heterogeneous feature spaces and improve interaction prediction.

IIFDTI (Cheng et al., 2022): Substructure interaction features are obtained through a bidirectional encoder-decoder framework. Drug features are extracted using a GNN, while protein features are extracted using a CNN. The features are then combined and passed through fully connected layers to predict DTIs.

TransformerCPI (Chen et al., 2020): The model identifies key interaction features by applying deconvolution to highlight critical regions in protein sequences and compound atoms. A label inversion experiment is also introduced to validate the model.

MolTrans (Huang et al., 2021): The FCS mining module first extracts

substructure sequences from drug and protein data using a specialized decomposition algorithm. These sequences are embedded through an enhanced Transformer module. Drug and protein substructures are paired, and higher-order interactions are captured with a CNN layer. The decoder then outputs interaction probabilities.

BINDTI (Peng et al., 2024): Drug features are encoded from SMILES strings into 2D molecular graphs using a GCN. Protein features are generated from amino acid sequences via the ACMIX model, which combines self-attention and convolution. A bidirectional interaction network fuses drug and protein features with multi-head attention. Finally, an MLP classifies unknown drug-target pairs based on the fused features.

4.3. Evaluation metrics

Model performance was evaluated using AUROC and AUPRC as the primary metrics (Heidari & Iosifidis, 2021; Peng et al., 2021; Zhao et al., 2020). In addition, standard categorical metrics including Accuracy, Sensitivity, Specificity, and F1 score were calculated, defined as follows:

AUROC: AUROC evaluates binary classification performance by calculating the area under the ROC curve. This metric reflects the balance between true positive and false positive rates. Higher AUROC values indicate better overall discrimination between classes.

$$FPR = \frac{FP}{FP + TN} \quad (39)$$

$$TPR = \frac{TP}{TP + FN} \quad (40)$$

AUPRC: AUPRC assesses binary classification performance by focusing on precision and recall for the positive class. This metric provides more informative results than AUROC, especially when class distributions are imbalanced.

$$Recall = \frac{TP}{TP + FN} \quad (41)$$

$$Precision = \frac{TP}{TP + FP} \quad (42)$$

Accuracy: Accuracy measures the proportion of correct predictions in classification tasks. It works well for balanced data but is less sensitive on imbalanced datasets. It is formally expressed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (43)$$

Sensitivity: Sensitivity, or recall, measures how accurately positive cases are identified, calculated as the ratio of true positives to all actual positives. It is defined as:

$$Sensitivity = Recall = \frac{TP}{TP + FN} \quad (44)$$

Specificity: Specificity measures the ability to correctly identify negative cases, defined as the ratio of true negatives to all actual negatives. The definition is given by:

$$Specificity = \frac{TN}{TN + FP} \quad (45)$$

F1: F1 score is the harmonic mean of precision and recall, balancing positive detection and prediction consistency. It is formally expressed as:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (46)$$

4.4. Comparison of LDM-DTI and baseline methods

In this section, model performance was assessed using five

evaluation metrics across the BindingDB, BioSNAP, *Human*, and *C. elegans* datasets. Comparisons were conducted with traditional machine learning methods and recent state-of-the-art baselines. Official implementations and reported optimal hyperparameters were adopted for each approach. The model with highest AUROC on validation set was selected. Corresponding AUROC and AUPRC scores are presented in Tables 2 and Table 3. Additional metrics, including Sensitivity, Specificity, Accuracy, and F1 score, are reported in Supplementary Tables S1-S4.

Table 3 provides a comparative analysis of the performance of LDM-DTI on the BindingDB and BioSNAP datasets. On the BindingDB dataset, LDM-DTI attained superior results across all evaluation metrics, with an AUROC of 0.960, AUPRC of 0.945, accuracy of 0.904, and F1 score of 0.902, outperforming all baseline models. The BINDTI model ranked second in overall performance. LDM-DTI outperformed the traditional RF model, with improvements of 1.8% in AUROC, 2.2% in AUPRC, 3.3% in Accuracy, and 5.8% in F1 score. Compared with the state-of-the-art IIFDTI model, gains of 3.6%, 4.4%, 7.1%, and 9.2% were observed for the same metrics. On the BioSNAP dataset, LDM-DTI achieved the highest predictive performance. It reached an AUROC of 0.908, AUPRC of 0.906, accuracy of 0.844 and F1 score of 0.839. Relative to the SVM model, improvements of 8.9% in AUROC, 6.7% in AUPRC, 9.4% in Accuracy, and 1.2% in F1 score were recorded. When compared with MolTrans, performance increased by 2.9%, 2.4%, 4.3%, and 3.7%, respectively. These improvements result from the combined use of GCN and EGNN, which capture multi-level structural features of drug molecules. This design allows the model to represent both local topological information and three-dimensional molecular geometry, enhancing predictive accuracy and structural understanding.

Table 4 summarizes LDM-DTI performance on the *Human* and *C. elegans* datasets. On the *Human* dataset, LDM-DTI achieved the highest results for all metrics. It reached AUROC of 0.981, AUPRC of 0.976, Accuracy of 0.946, and F1 score of 0.936. BINDTI ranked second. Compared with the traditional SVM, LDM-DTI improved AUROC by 6.8%, AUPRC by 7.1%, Accuracy by 10.8%, and F1 score by 12.5%. Relative to TransformerCPI, gains of 2.5%, 3.3%, 4.5%, and 4.6% were observed. On the *C. elegans* dataset, LDM-DTI again outperformed all models. It reached AUROC and AUPRC values of 0.988 and 0.986, with Accuracy of 0.960 and F1 score of 0.956. Compared with KNN, improvements of 7.6%, 9.0%, 9.8%, and 9.5% were recorded. Performance enhancements over GraphDTI were 2.9%, 3.5%, 4.8%, and 4.6%,

Table 3

Results for the BindingDB, BioSNAP datasets and comparison with the baseline method.

Methods	BindingDB AUROC	AUPRC	BioSNAP AUROC	AUPRC
SVM	0.904 ± 0.002	0.865 ± 0.001	0.819 ± 0.045	0.839 ± 0.038
RF	0.942 ± 0.001	0.923 ± 0.001	0.857 ± 0.009	0.872 ± 0.006
KNN	0.895 ± 0.001	0.865 ± 0.002	0.842 ± 0.003	0.805 ± 0.003
LR	0.916 ± 0.003	0.884 ± 0.002	0.821 ± 0.004	0.796 ± 0.002
GraphDTA	0.944 ± 0.004	0.925 ± 0.006	0.871 ± 0.004	0.870 ± 0.006
DrugBAN	0.952 ± 0.001	0.936 ± 0.001	0.902 ± 0.001	0.905 ± 0.002
IIFDTI	0.924 ± 0.002	0.901 ± 0.004	0.895 ± 0.003	0.898 ± 0.005
TransformerCPI	0.895 ± 0.002	0.861 ± 0.003	0.843 ± 0.008	0.859 ± 0.007
MolTrans	0.935 ± 0.003	0.901 ± 0.004	0.879 ± 0.006	0.882 ± 0.006
BINDTI	0.956 ± 0.001	0.942 ± 0.002	0.895 ± 0.003	0.894 ± 0.003
LDM-DTI	0.960 ± 0.002	0.945 ± 0.002	0.908 ± 0.003	0.906 ± 0.003

Table 4

Results for the *Human*, *C. elegans* dataset and comparison with the baseline method.

Methods	<i>Human</i> AUROC	AUPRC	<i>C.elegans</i> AUROC	AUPRC
SVM	0.913 ± 0.012	0.905 ± 0.001	0.925 ± 0.009	0.907 ± 0.002
RF	0.939 ± 0.009	0.927 ± 0.001	0.935 ± 0.009	0.931 ± 0.001
KNN	0.915 ± 0.002	0.902 ± 0.002	0.912 ± 0.003	0.896 ± 0.002
LR	0.895 ± 0.003	0.862 ± 0.001	0.892 ± 0.002	0.883 ± 0.003
GraphDTA	0.965 ± 0.003	0.955 ± 0.003	0.959 ± 0.004	0.951 ± 0.002
DrugBAN	0.979 ± 0.001	0.969 ± 0.005	0.981 ± 0.002	0.978 ± 0.002
IIFDTI	0.977 ± 0.003	0.967 ± 0.027	0.984 ± 0.002	0.979 ± 0.002
TransformerCPI	0.956 ± 0.005	0.943 ± 0.002	0.975 ± 0.003	0.975 ± 0.002
MolTrans	0.979 ± 0.003	0.975 ± 0.002	0.982 ± 0.002	0.979 ± 0.002
BINDTI	0.980 ± 0.001	0.975 ± 0.004	0.985 ± 0.002	0.984 ± 0.002
LDM-DTI	0.981 ± 0.001	0.976 ± 0.002	0.988 ± 0.002	0.986 ± 0.002

respectively. These findings highlight the generalization ability of LDM-DTI across species. The dynamic weighted cross-attention mechanism also improves robustness against diverse protein structures, ensuring

stable and accurate predictions in different biological contexts.

In addition to the quantitative evaluations, Fig. 2. provides a boxplot-based comparative analysis of predictive accuracy for LDM-DTI and baseline models across the four benchmark datasets. The figure visually illustrates performance stability under five-fold cross-validation, capturing both central tendency and variance. As shown, LDM-DTI consistently attains the highest median accuracy across all datasets, while also demonstrating the narrowest interquartile ranges. These results indicate not only superior average predictive performance but also enhanced robustness and reduced variability, surpassing traditional machine learning methods such as SVM and RF, as well as advanced deep learning frameworks including MolTrans and BINDTI. The consistently stable results observed on BindingDB, BioSNAP, *Human*, and *C.elegans* confirm the strong generalization capability of LDM-DTI across heterogeneous data distributions and species. Collectively, these findings underscore the effectiveness of the proposed framework in integrating semantic, structural, and attentional representations to enable reliable and robust DTI prediction.

This study performed t-tests (Wang et al., 2025) to assess the statistical significance of performance differences between LDM-DTI and each baseline model, using AUROC values obtained from five independent runs on each dataset. The resulting t- and p-values are summarized in Supplementary Table S5 and visualized as a heatmap in Fig. 3. LDM-DTI exhibits statistically significant improvements ($p < 0.01$) over the majority of baselines across all datasets. For instance, although TransformerCPI yields high t-values on BindingDB ($t = 65.97$) and BioSNAP ($t = 17.01$), its overall predictive accuracy remains substantially lower than that of LDM-DTI. Similarly, while MolTrans performs competitively

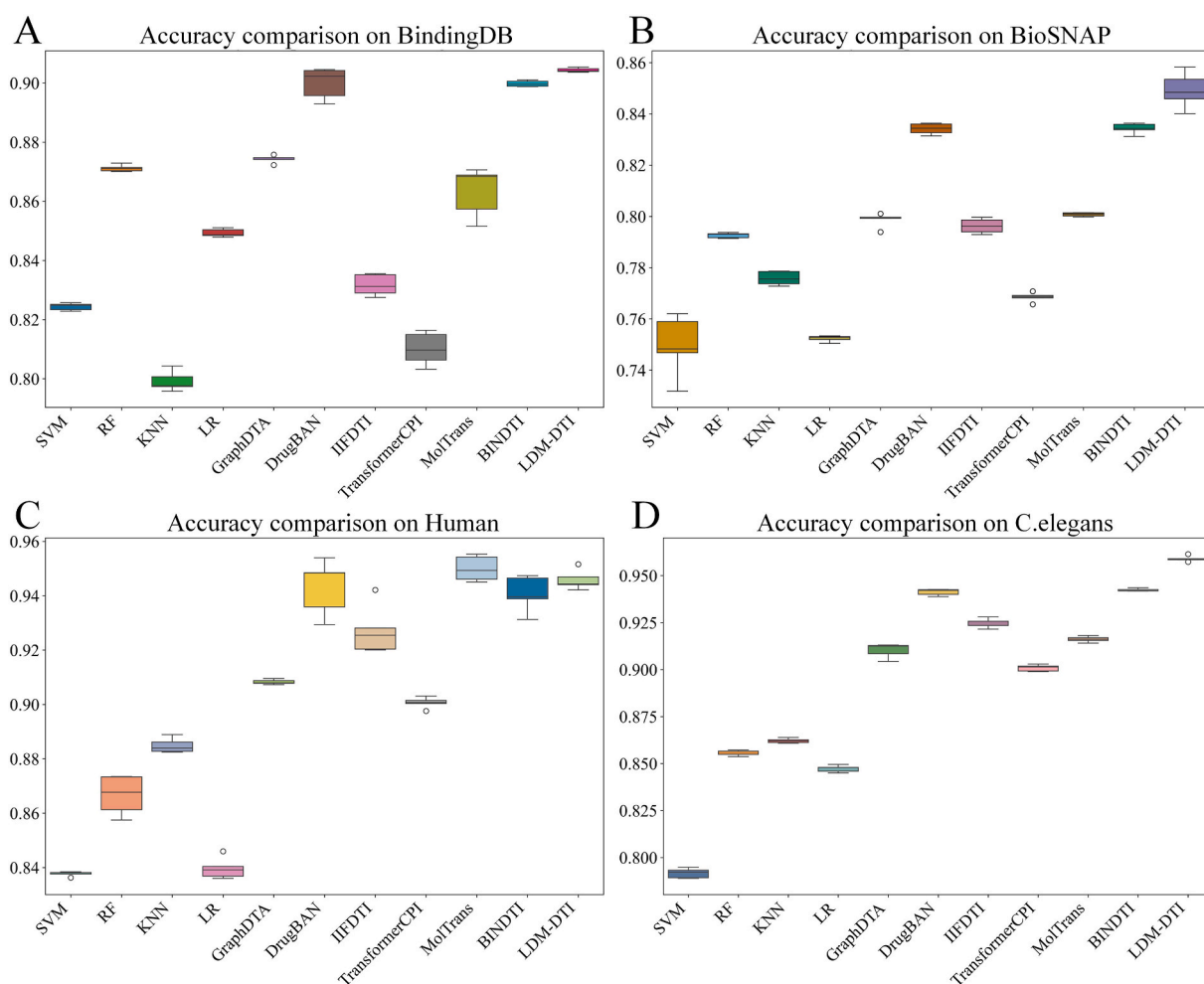


Fig. 2. Accuracy comparison of LDM-DTI and baseline models across four benchmark datasets. (A) BindingDB. (B) BioSNAP. (C) *Human*. (D) *C.elegans*.

on BioSNAP ($t = 16.67$, $p < 0.001$), it underperforms on the *Human* dataset, where LDM-DTI maintains higher t-values and lower variance. Notably, on the *C. elegans* dataset, LDM-DTI achieves strong significance despite limited sample size, reflecting its robustness to cross-species heterogeneity. These statistical analyses indicate that the performance gains of LDM-DTI are both stable and reliable, consistently exceeding those of benchmark models across diverse datasets and biological conditions. This underscores its outstanding generalization capacity and adaptability, providing strong statistical evidence for its effectiveness in predicting diverse DTIs.

To confirm the predictive advantage of LDM-DTI, additional statistical testing was conducted. A two-tailed t -test (Wang et al., 2025) was applied to compare LDM-DTI with other classifiers on the BindingDB dataset, using a significance threshold of $p < 0.05$. The test outcomes are summarized in Table 4. In addition, 95% confidence intervals were calculated to assess the statistical significance of the prediction performance between LDM-DTI and ten baseline models. The results on the BindingDB dataset are shown in Table 5, while those for other datasets are provided in Supplementary Section S6. The outcomes of significance testing are summarized in Supplementary Section S7.

Table 5 indicates that LDM-DTI consistently achieves superior predictive performance compared to all baseline classifiers, including SVM, RF, KNN, LR, GraphDTA, DrugBAN, IIFDTI, TransformerCPI, MolTrans, and BINDTI. Statistically significant differences are observed across all evaluation metrics. Specifically, the p-values for AUPRC range from $1.39\text{E-}08$ to $4.15\text{E-}02$, for Accuracy from $2.49\text{E-}08$ to $6.69\text{E-}02$, and for F1-score from $2.07\text{E-}09$ to $3.58\text{E-}01$. Most of these p-values are below the commonly accepted significance level of $\alpha = 0.05$, indicating that LDM-DTI outperforms competing methods in the majority of cases and demonstrates enhanced effectiveness in DTI prediction. These results suggest that the model captures intricate patterns in drug-protein interactions, leading to stable and accurate predictions. The statistical evidence further supports the consistency of these improvements across different datasets, demonstrating reliability in DTI prediction.

As summarized in Table 6, LDM-DTI demonstrated the highest predictive accuracy among all compared methods. On the BindingDB dataset, both the lower and upper bounds of its confidence intervals for AUPRC, Accuracy, and F1 were consistently above those of SVM, RF, KNN, LR, GraphDTA, DrugBAN, IIFDTI, TransformerCPI, MolTrans, and BINDTI. Specifically, the lower bounds for AUPRC exceeded those of competing models by 0.5%-8.44%, and the upper bounds by 0.2%-8.08%. Accuracy showed similar gains, with increases of 0.36%-7.8% and 0.70%-8.19% for the lower and upper bounds, respectively. The relatively narrow confidence intervals indicate minimal performance fluctuations across repeated trials, reflecting the consistent behavior of

Table 5

P-value comparison between LDM-DTI and other classifiers on AUROC, accuracy, and F1 metrics on the BindingDB dataset.

Methods	P-value(AUPRC)	P-value(Acc)	P-value(F1)
SVM	1.39E-08	3.92E-08	2.07E-09
RF	4.14E-06	8.90E-06	1.28E-07
KNN	1.68E-07	2.49E-08	5.65E-08
LR	1.07E-06	3.06E-06	1.51E-06
GraphDTA	2.51E-04	2.05E-05	6.13E-03
DrugBAN	4.05E-01	6.69E-02	1.15E-02
IIFDTI	1.38E-06	2.32E-07	8.21E-10
TransformerCPI	1.22E-07	3.64E-06	1.39E-07
MolTrans	5.21E-05	1.65E-03	7.26E-07
BINDTI	4.15E-02	7.03E-03	3.58E-01

Table 6

Comparison of confidence intervals between dataset BindingDB and other classifiers in LDM-DTI.

Confidence interval Classifier	AUPRC(%)		Acc(%)		F1(%)	
	Lower	Upper	Lower	Upper	Lower	Upper
SVM	86.46	86.63	82.34	82.55	78.30	81.12
RF	92.15	92.31	86.96	87.22	84.19	85.02
KNN	86.25	86.83	79.39	79.88	79.45	89.44
LR	88.08	88.60	84.76	85.11	83.97	84.05
GraphDTA	91.43	92.65	87.34	87.54	87.02	92.53
DrugBAN	94.44	94.70	89.78	90.45	90.04	94.38
IIFDTI	89.75	90.39	82.96	83.50	80.63	91.79
TransformerCPI	85.97	86.55	80.64	81.81	77.73	89.26
MolTrans	89.55	90.73	85.19	87.69	83.20	94.32
BINDTI	93.91	94.49	89.78	90.04	89.67	93.82
LDM-DTI	94.41	94.63	90.14	90.74	90.09	94.20

the model. These results imply that LDM-DTI effectively identifies intricate interaction patterns between drugs and proteins, supporting dependable predictive outcomes.

4.5. Ablation study

Ablation experiments were performed on the BioSNAP dataset to investigate how each major component contributes to the overall performance of the LDM-DTI framework. Four simplified versions of the model were constructed for comparison with the complete architecture. To explore the combined effect of the GCN and EGNN modules in drug feature extraction, two variants were implemented: one with the GCN module removed while retaining only EGNN (denoted as Without GCN), and the other with EGNN removed and only GCN preserved (Without EGNN). For protein representation, the DCNN was replaced by a standard CNN (Without DCNN) to test whether adaptive filtering improves feature extraction. Lastly, to evaluate the effectiveness of the dynamically weighted attention mechanism in cross-modal fusion, this module was replaced with a conventional multi-head attention mechanism (Without DIAM), enabling assessment of its contribution to interaction modeling.

In this study, the experimental results were further examined using ROC and PR curve analyses, as shown in Fig. 4. A marked decline in performance was observed when the GCN module was removed and only the EGNN was retained for drug representation, with AUROC dropping from 0.908 to 0.765 and AUPRC from 0.906 to 0.714. These results demonstrate that the GCN plays a vital role in learning drug-related features, as its removal causes a clear loss of predictive accuracy. Conversely, when EGNN was omitted and only the GCN component was preserved, AUROC and AUPRC decreased to 0.796 and 0.735, respectively. Although the performance still deteriorated, the reduction was slightly smaller than that observed without the GCN, suggesting that EGNN also contributes meaningfully to the representation of drug molecules. When the DCNN module is replaced by a standard CNN for protein feature extraction, the AUROC and AUPRC scores drop to 0.807

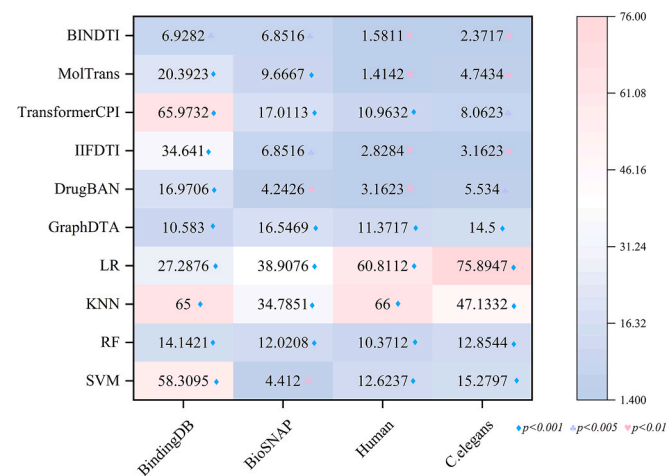


Fig. 3. Statistical tests of five randomized experiments on the BindingDB, BioSNAP, *Human* and *C. elegans* datasets.

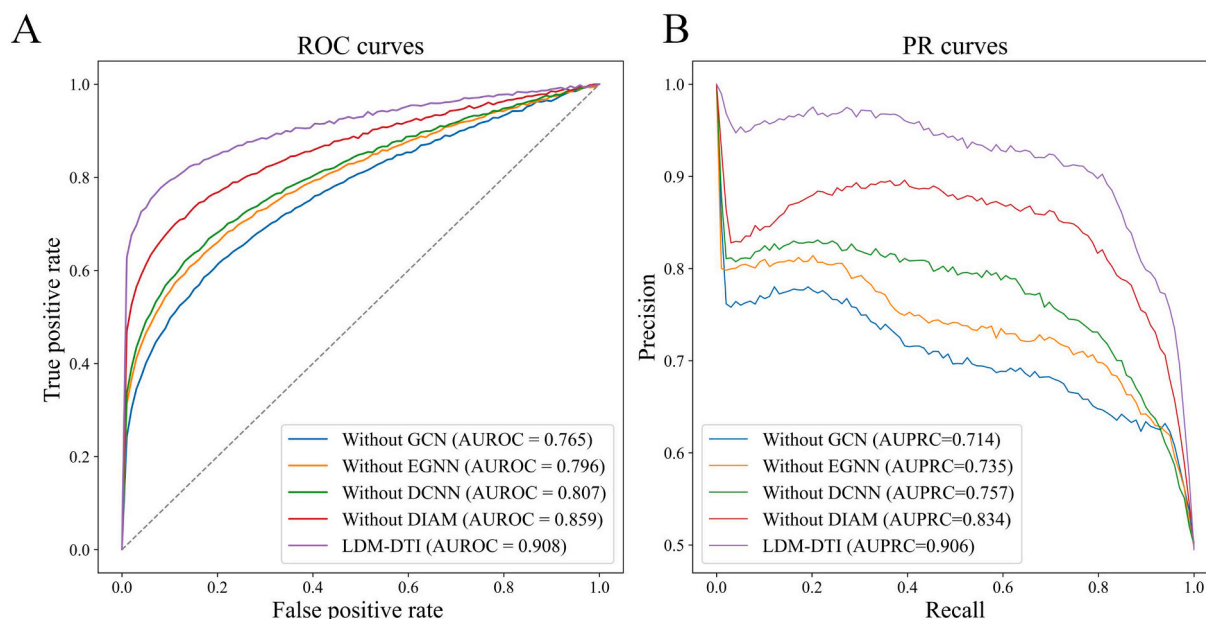


Fig. 4. Performance comparison of ablation experiments on the BioSNAP. (A) is ROC curve and (B) is PR curve.

and 0.757. This reduction demonstrates that the DCNN module enables capturing key protein features more effectively than a conventional CNN. Owing to its adaptive filtering mechanism, the dynamic convolution network can adjust its kernels in response to the specific sequence context, allowing it to model local residue dependencies with greater precision. Replacing the DIAM module with a conventional multi-head attention mechanism in the feature fusion stage led to declines in AUROC to 0.859 and AUPRC to 0.834.

Among the ablation variants, removing the GCN module caused the greatest performance reduction, with AUROC dropping by 14.3% and AUPRC by 19.2%. Excluding the EGNN module also resulted in notable declines of 11.2% in AUROC and 17.1% in AUPRC. These findings demonstrate the necessity of combining topological and geometric structural information through the GCN and EGNN modules to achieve effective molecular representation. Although the model still performed reasonably well without DIAM, the observed decrease emphasizes the significance of dynamic weighting in optimizing feature integration. Finally, removing the DCNN module reduced AUROC by about 10% and AUPRC by 15%, further confirming its role in capturing local structural dependencies within protein sequences. In general, each component of the framework contributes to improving the predictive performance of the model.

4.6. Interpretability analysis

This study investigates how the attention mechanism highlights critical residues involved in drug-protein binding. When drug SMILES strings and protein sequences are input, the attention module generates a matrix emphasizing the most relevant residues for binding. Molecular docking was subsequently performed using AutoDock to validate the predicted binding sites of drug-protein complexes. By comparing docking outcomes with the residues identified by the model, we confirmed that the model effectively captures key amino acids involved in molecular interactions. The visualization combining predicted attention weights and binding pockets is displayed in Fig. 5.

Fig. 5A depicts the KDR protein complexed with the ligand sorafenib (PDB ID: 3WZE (Okamoto et al., 2015)). As shown in Fig. 5A, the overall structure of BRAF appears in light purple, whereas the bound molecule of sorafenib is illustrated in yellow near the kinase active site. A closer inspection reveals two residues within the kinase domain that interact

directly with the drug. Their names are labeled in the figure, together with the measured distances in angstroms (Å). BRAF functions as a tyrosine kinase essential for regulating cellular proliferation and growth. Sorafenib, an orally available small-molecule inhibitor, acts on multiple tyrosine kinases and is clinically used for hepatocellular carcinoma (HCC), renal cell carcinoma (RCC), and thyroid cancer. Sorafenib acts by competitively occupying the ATP-binding pocket of the BRAFV600E mutant, thereby suppressing its kinase activity and, consequently, tumor cell proliferation. When the drug binds to BRAF, several stabilizing forces come into play, including hydrogen bonds and hydrophobic contacts. Among the surrounding residues, ARG-880, a positively charged amino acid, appears to interact with hydrophilic or negatively charged groups on Sorafenib, which helps reinforce the stability of the drug-protein complex. In addition, SER-884, a polar residue within the kinase domain, seems to strengthen the association further through hydrogen bonding with specific functional groups of Sorafenib.

As shown in Fig. 5B, the EGFR tyrosine kinase domain appears in green, while the inhibitor Gefitinib is colored cyan. A closer view reveals the interaction between Gefitinib and residue ASP-813, highlighted in blue. The measured distance of 2.0 Å suggests a strong binding, which appears to involve hydrogen bonding, with ASP-813 playing a key role in stabilizing the drug-protein complex. EGFR, a membrane-bound tyrosine kinase receptor, plays key roles in regulating cellular processes such as proliferation, differentiation, and survival. The binding of epidermal growth factor triggers downstream pathways, including the MAPK and PI3K/Akt cascades, thereby promoting cell growth. Gefitinib, approved for non-small cell lung cancer, targets the kinase domain of EGFR and shows particular efficacy in patients carrying mutations such as L858R and T790M. Within the kinase domain, the polar residue ASP-813 appears to interact closely with Gefitinib, engaging with its aromatic ring and nitrogen atoms through hydrogen bonds and electrostatic interactions. These contacts help strengthen the binding affinity of Gefitinib, suppressing EGFR kinase activity. Understanding the role of ASP-813 in drug binding provides valuable insights for the rational design of more selective inhibitors. Such structural optimization may also aid in overcoming drug resistance observed in certain EGFR mutant phenotypes.

To further clarify the internal mechanisms and feature learning process of the model, module-specific weight visualizations were generated. Fig. 6 presents the EGNN, GCN, Transformer, DCNN, and

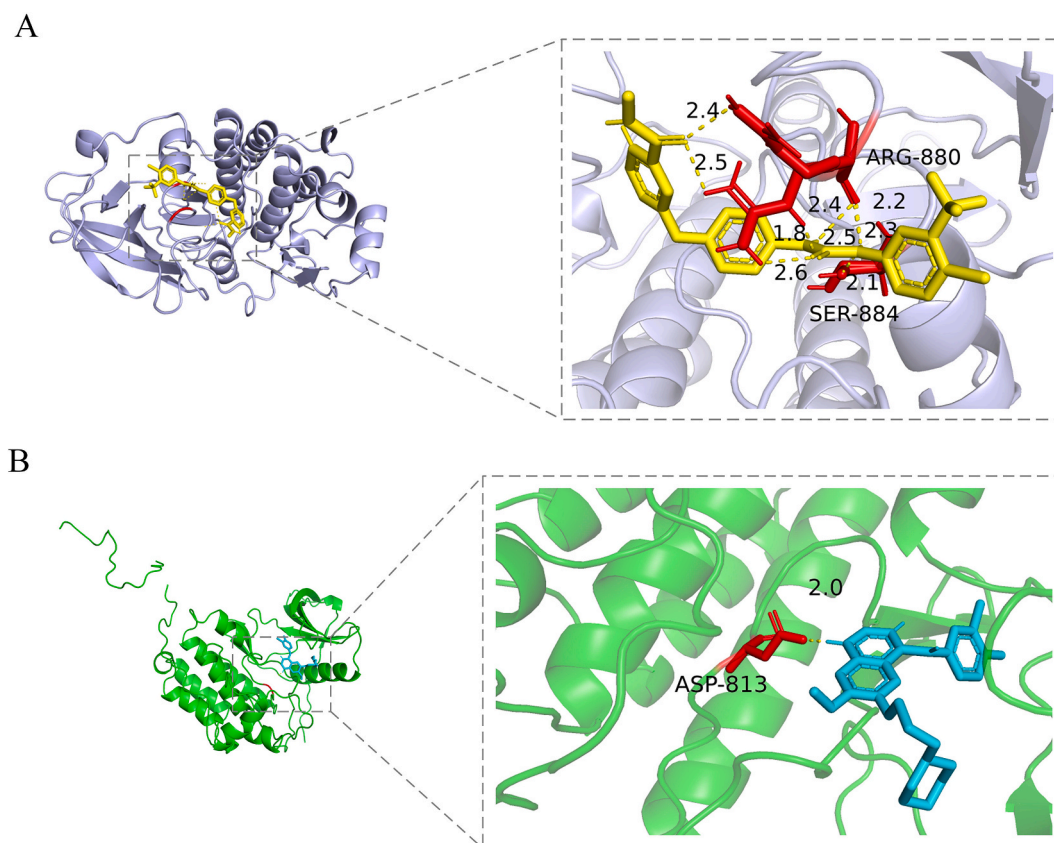


Fig. 5. Visualization of predicted protein attention weights alongside binding pockets. (A) Docking diagram of Sorafenib with the structural domain of BRAF kinase (PDB ID: 3WZE). (B) Schematic illustration of Gefitinib binding to EGFR kinase at key residue ASP-813 (PDB ID: 1 M17).

Dynamic Weight modules. In these visualizations, color intensity indicates attention to specific spatial features, with darker colors corresponding to higher weights and stronger influence on prediction outcomes. The weight distributions of the EGNN and GCN modules in the drug feature encoder show concentrated regions within the feature space. This indicates that the model effectively identifies atomic substructures important for drug activity while capturing molecular topology. In particular, the EGNN module improves spatial structure representation by integrating geometric information, enhancing the description of local functional groups in drugs. By comparison, the Transformer and DCNN modules in the protein feature encoder display distinct preferences in modeling sequence features. The Transformer module forms global connections among distant residues through self-attention. Its weight distribution is relatively dispersed but covers a wide range, highlighting its effectiveness in modeling long-range dependencies. In contrast, the DCNN module shows a more concentrated pattern, using multi-scale convolutional structures to extract local

fragment features. This supports the capture of important sequence patterns, including conserved regions and structural domains. During feature fusion, the dynamic weighting module combines these representations through adaptive weighting. The visualizations show that attention is strongly focused within the region marked by the red box near the sequence terminus. This suggests that the module assigns higher weights to informative features, emphasizing key structural elements.

To further clarify how the model makes predictions in DTI tasks, interpretability analysis was performed using the SHapley Additive exPlanations (SHAP) method (Lundberg et al., 2020). Fig. 7 depicts how drug and protein features influence model predictions. Each row represents a specific feature dimension (e.g., D52 for the 52nd drug feature and P87 for the 87th protein feature), while the horizontal axis denotes the SHAP value, reflecting both the strength and direction of each feature's impact on the prediction. Dot colors reflect the actual feature values: red indicates high values, blue represents low values, and purple

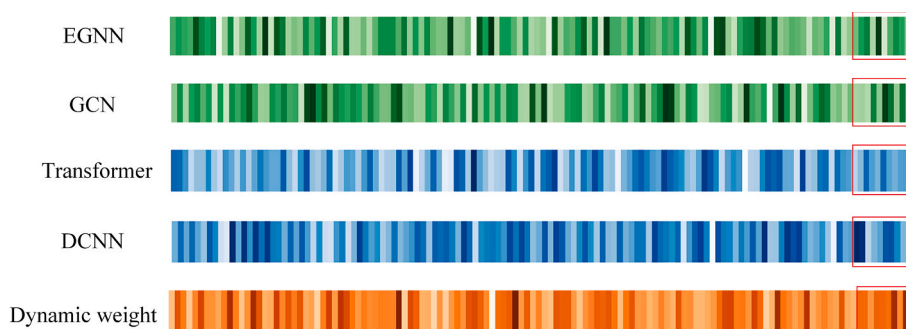


Fig. 6. Results of visualizing the output weights of the modules.

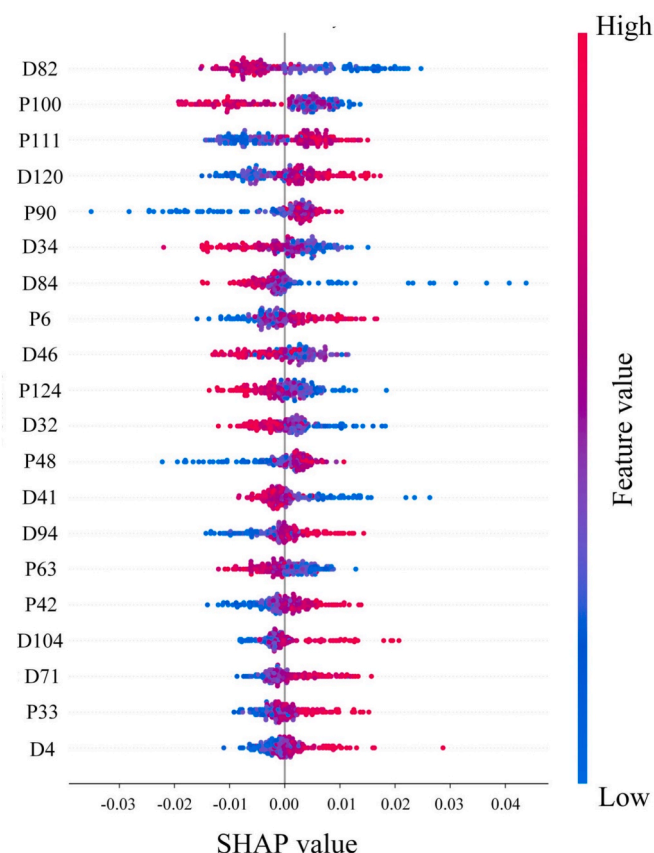


Fig. 7. Visualization of the SHAP values of the features in DTI prediction results.

denotes intermediate levels. Features such as D124, D87, and P110 exhibit consistently large positive SHAP values across samples, suggesting strong contributions to the predictive outcomes. In contrast, features such as D6 and P58 show negative SHAP values at high actual values, suggesting a potential inhibitory effect on prediction performance. This analysis provides insights into the dependence on specific feature dimensions and identifies the drug and protein features that most strongly influence predictions. Such interpretability not only enhances model transparency but also provides theoretical guidance for downstream tasks such as feature selection, drug optimization, and target identification.

4.7. Case study

To further demonstrate effectiveness, two case studies were conducted, examining drug and target perspectives separately. Two representative drugs, DB01050 (Ibuprofen) and DB01254 (Dasatinib), and two representative targets, P21554 (Cannabinoid receptor 1) and P35354 (Prostaglandin G/H synthase 2), were chosen. The model was trained on the BindingDB dataset, with all interaction records involving these four entities removed beforehand to maintain objective evaluation. After training, candidate targets for each drug and candidate drugs for each target were ranked according to the predicted interaction scores. The top-ranked predictions were compared with known interactions in the database to evaluate model performance.

To examine the potential biological relevance of the predicted interactions, a network was constructed using the NR dataset (Shi et al., 2019). Eighty drug-target pairs from the model output were used to build four drug-target interaction sub-networks (DTISs), as shown in Fig. 8. Among these pairs, 75 matched known interactions in the reference database, resulting in an accuracy of 93.75%. This high level of

agreement indicates that the model can generalize to unseen drugs and targets in a zero-shot prediction setting. However, five interactions were not correctly identified, suggesting that the model may struggle with compounds that have complex structures or unclear biological functions.

For instance, the model incorrectly predicted the lipid analog DB16495 (oleic monoethanolamide, OEA) as a ligand for CB1. CB1 participates in physiological processes including memory, appetite, mood regulation, and pain perception. OEA shares some functional overlap with these processes, but its main mechanism occurs through activation of PPAR- α rather than CB1. In addition, OEA can inhibit ceramidase and influence sphingolipid metabolism. This misprediction might result from chemical structural similarities or incomplete database information about interactions between this compound class and CB1.

Dasatinib was correctly associated with several tyrosine kinase targets, including ABL1 and SRC, which matches its known broad-spectrum anticancer activity. Ibuprofen was linked to its established target COX-2 and also predicted to interact with estrogen receptor alpha, indicating that this NSAID class may have additional hormone-modulating effects. In the target-focused predictions, COX-2 was connected to several well-characterized NSAIDs, such as Nimesulide and Celecoxib. The model also predicted a possible interaction with Birinapant (DB09363), a modulator of the apoptosis pathway. This result suggests potential overlap between anti-inflammatory and anticancer mechanisms, highlighting the capacity to reveal novel DTIs and support drug repurposing. Some prediction mismatches may exist due to incomplete database records. Certain DTIs are still being tested experimentally or clinically and are not yet included in standard resources like BindingDB.

5. Conclusion

DTI prediction plays a critical role in drug discovery and development, facilitating faster identification of new therapeutic compounds and reducing associated costs. To address challenges in this area, a new end-to-end deep learning framework, LDM-DTI, was developed. ChemBERTa is leveraged to encode drug SMILES sequences, while ProtBERT encodes protein sequences, eliminating reliance on manual feature engineering and enabling automatic extraction of high-level representations from large-scale data. For drug feature encoding, GCN and EGNN are combined to aggregate and propagate node-level information according to the molecular graph structure. The use of EGNs, which maintain equivariance under rotational and translational transformations, enables the model to capture complex spatial relationships among atoms, chemical bonds, and cross-modal features. This approach improves the structural representation capability of the drug encoder, especially for large molecular datasets. Additionally, a new module combining a DCNN with a Transformer was developed to encode protein features. DCNNs are well suited for processing sequential protein data and can capture long-range amino acid interactions relevant to secondary structure prediction. Transformers, with their self-attention mechanisms, are effective at modeling global dependencies. Combining these two architectures improves the capacity of the model to generalize to previously unseen drugs and proteins. In addition, a dynamic feature interaction module is introduced, which adjusts weights based on the characteristics of the input data. Compared with static weighting methods, this approach adapts more effectively to complex and variable DTI data, enhancing generalization and reducing overfitting risk. The performance of the proposed LDM-DTI model was assessed against ten baseline methods across four benchmark datasets: BindingDB, BioSNAP, Human, and C.elegans. LDM-DTI consistently achieved superior results in metrics including AUROC and AUPRC, with statistical significance confirmed via t-tests. Its predictive capability was further supported by multi-dimensional interpretability analyses. In a case study, the model correctly predicted 75 of 80 drug-target pairs, yielding an accuracy of 93.75%. Overall, LDM-DTI demonstrates robust

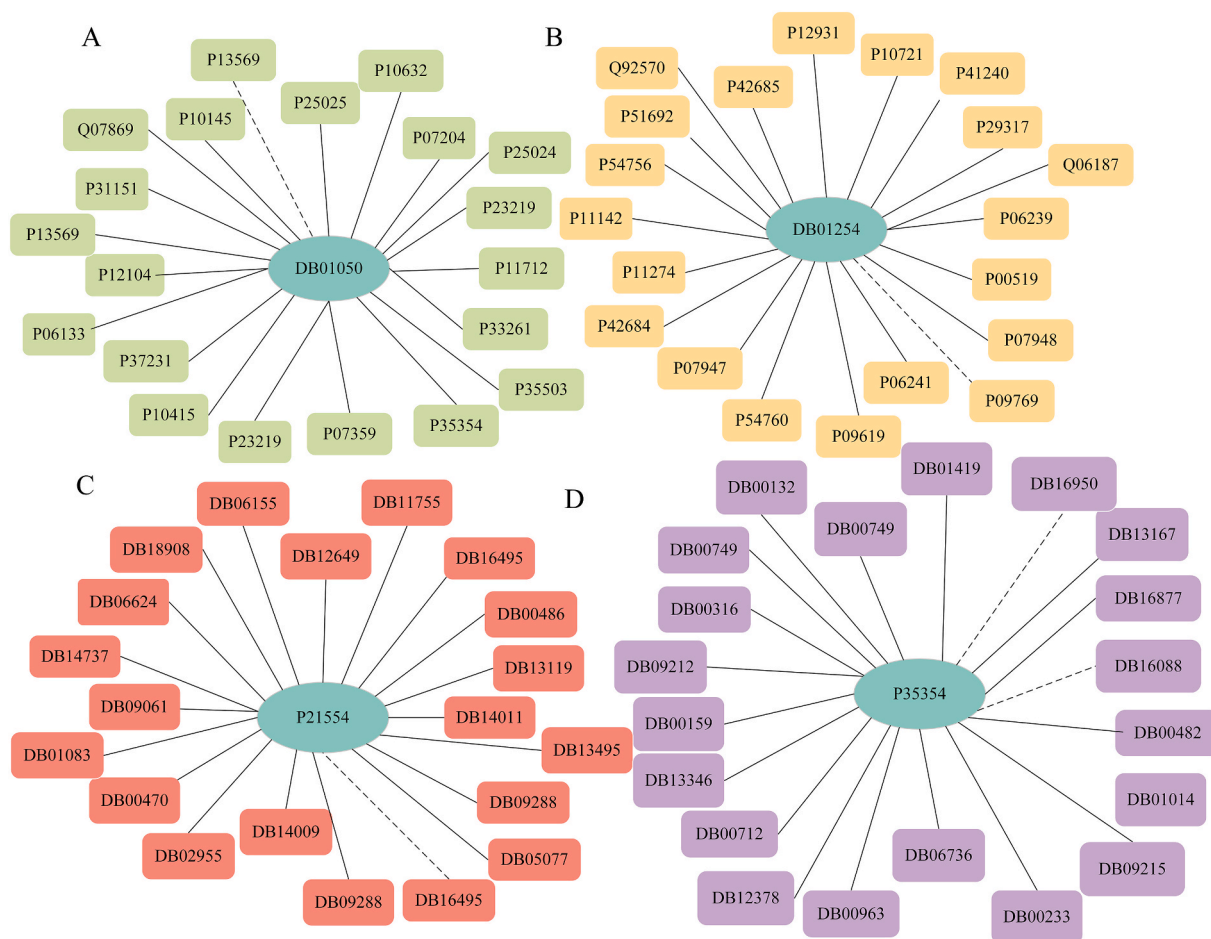


Fig. 8. Four DTI sub-networks (DTIS). Nodes indicate drugs or targets, edges indicate predicted interactions, and solid lines indicate correct predictions and dashed lines indicate incorrect predictions. (A) Candidate target sub-network corresponding to Ibuprofen. (B) Candidate target sub-network corresponding to Dasatinib. (C) Candidate drug sub-network corresponding to Cannabinoid receptor 1. (D) Candidate drug sub-network corresponding to Prostaglandin G/H synthase 2.

performance in DTI prediction, benefiting from its novel algorithmic design and multimodal feature learning framework.

LDM-DTI exhibits strong predictive capability, but its computational demands present a notable limitation. The combination of GCNs, DCNN, and multi-head attention mechanisms imposes substantial computational costs, particularly when applied to large-scale datasets. In addition to efficiency considerations, practical implementation in drug discovery requires access to high-quality biochemical and structural data, as well as systematic experimental validation to ensure the reliability of predictions. Future work will focus on enhancing computational efficiency through architectural optimization, algorithmic refinement, and hardware acceleration. Efforts will also be directed toward integrating computational predictions with experimental strategies, thereby improving the scalability and real-world applicability of LDM-DTI in drug development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2026.131485>.

Data availability

The data that has been used is confidential.

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